Ross Systems and Controls

LAKE TOBESOFKEE

SN: 201-040

PROJECT # 201-040

DATE: 06/18/2009

OPERATOR INTERFACE

MANUAL



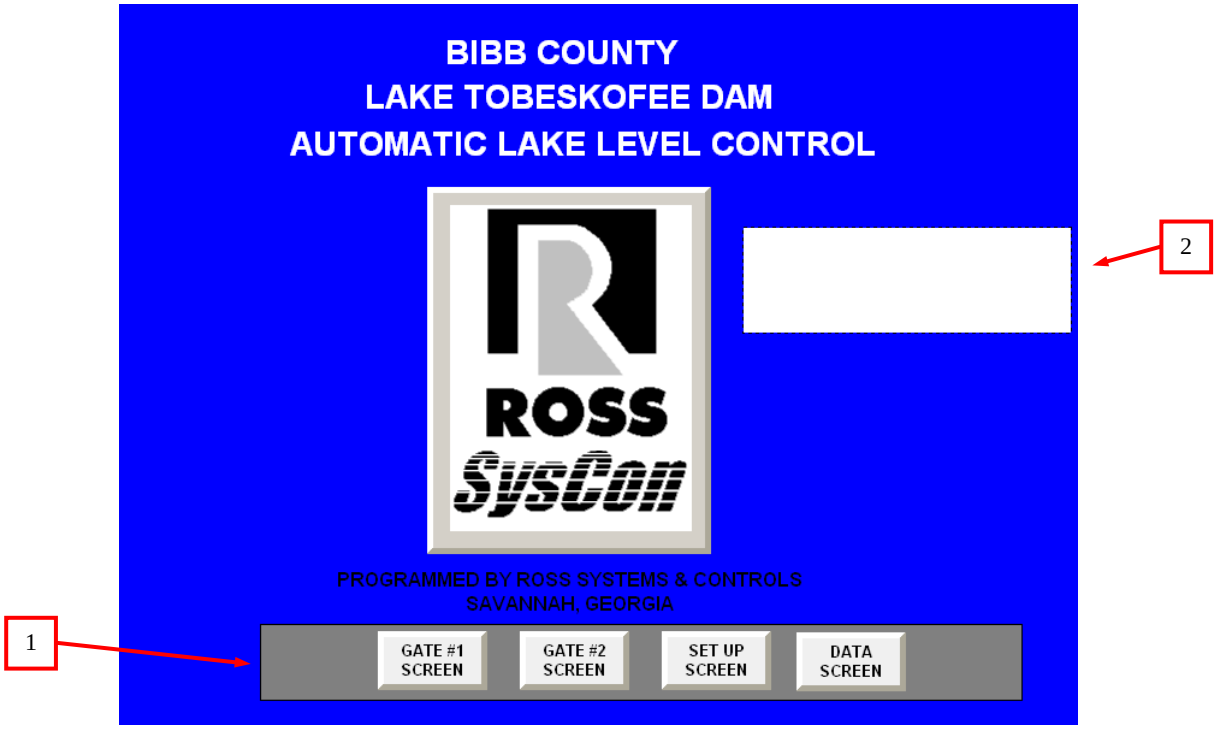
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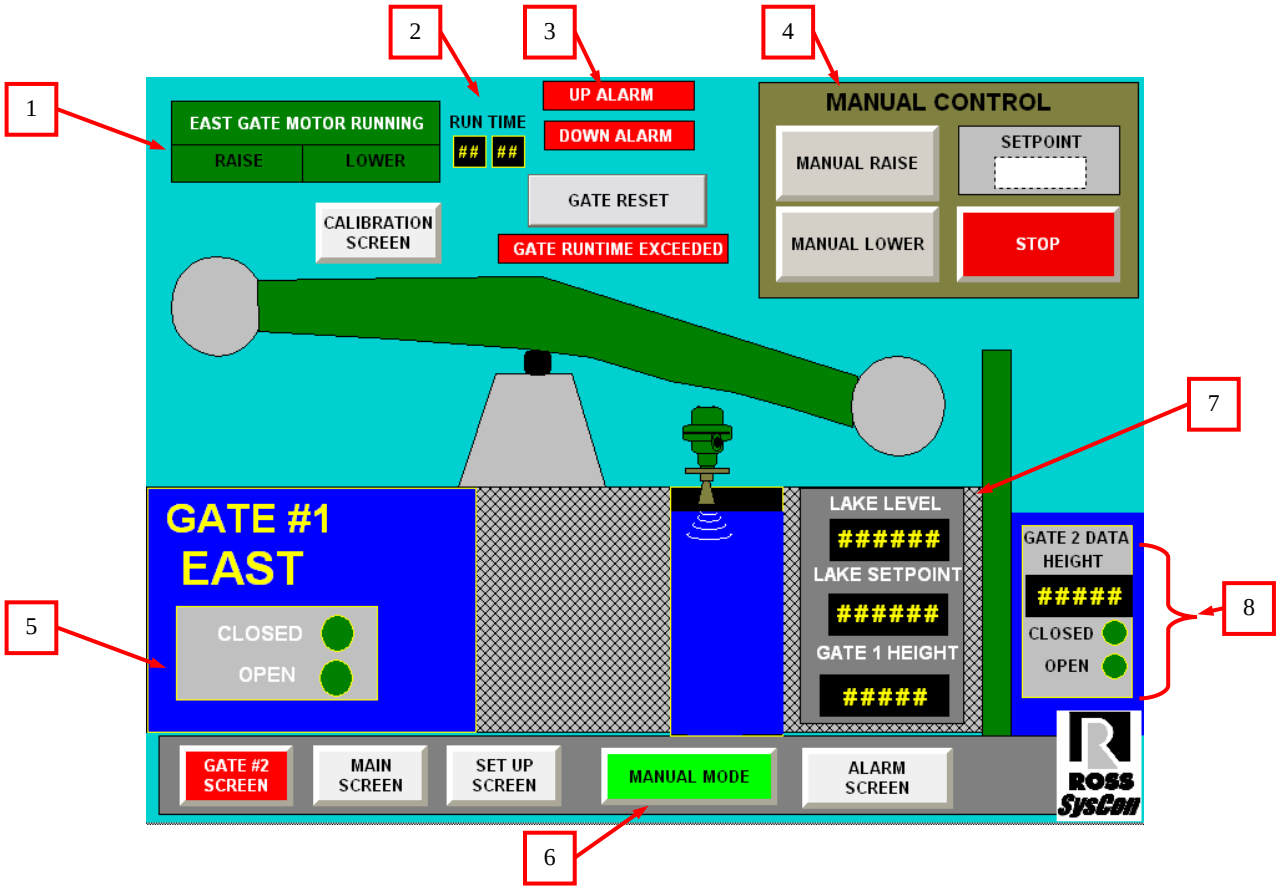
OPERATOR DESCRIPTION MAIN PLC PANEL

- MODE SWITCH: SELECTS BETWEEN AUTO AND MANUAL MODE. AUTO MODE IS FOR COMPUTER AND PLC CONTROL. MANUAL MODE ALLOWS OPERATOR TO OPERATE GATE WITHOUT PLC CONTROL.
- GATE #1 SWITCH: OPEN – SELECT OPEN TO RAISE GATE. MOTOR WILL RUN FOR PREDETERMINED RUN TIME BASED ON TIMER INSIDE PANEL. MAXIMUM RUN TIME IS xxxx BEFORE THE GATE STOPS AND A NEW COMMAND MUST BE GIVEN.
- CLOSE -SELECTING CLOSE WILL LOWER THE GATE FOR THE SAME AMOUNT OF TIME
- GATE #2 SWITCH: OPEN – SELECT OPEN TO RAISE GATE. MOTOR WILL RUN FOR PREDETERMINED RUN TIME BASED ON TIMER INSIDE PANEL (SAME AS GATE 1).
- CLOSE -SELECTING CLOSE WILL LOWER THE GATE FOR THE SAME AMOUNT OF TIME
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MAIN SCREEN

1) **SCREEN NAVIGATION BUTTONS:** ADVANCES OPERATOR TO SELECTED SCREEN WHEN DEPRESSED.
2) **MESSAGE BLOCK:** THIS BLOCK IS USED TO RELAY MESSAGES TO THE OPERATORS. TYPE IN MESSAGE AND PRESS ENTER.



EAST/ GATE 1 SCREEN

1). GATE DIRECTION INDICATION: ILLUMINATES THE DIRECTION OF THE GATE WHEN IN RUN MODE.

2). GATE RUN TIME: DISPLAYS THE TOTAL AMOUNT OF RUN TIME FOR GATE 1. RUN TIME LIMIT IS SET ON CALIBRATION PAGE.

3). GATE ALARMS: INDICATORS APPEAR WHEN GATE IS IN ALARM STATE. PRESS RESET TO CLEAR ALARM.

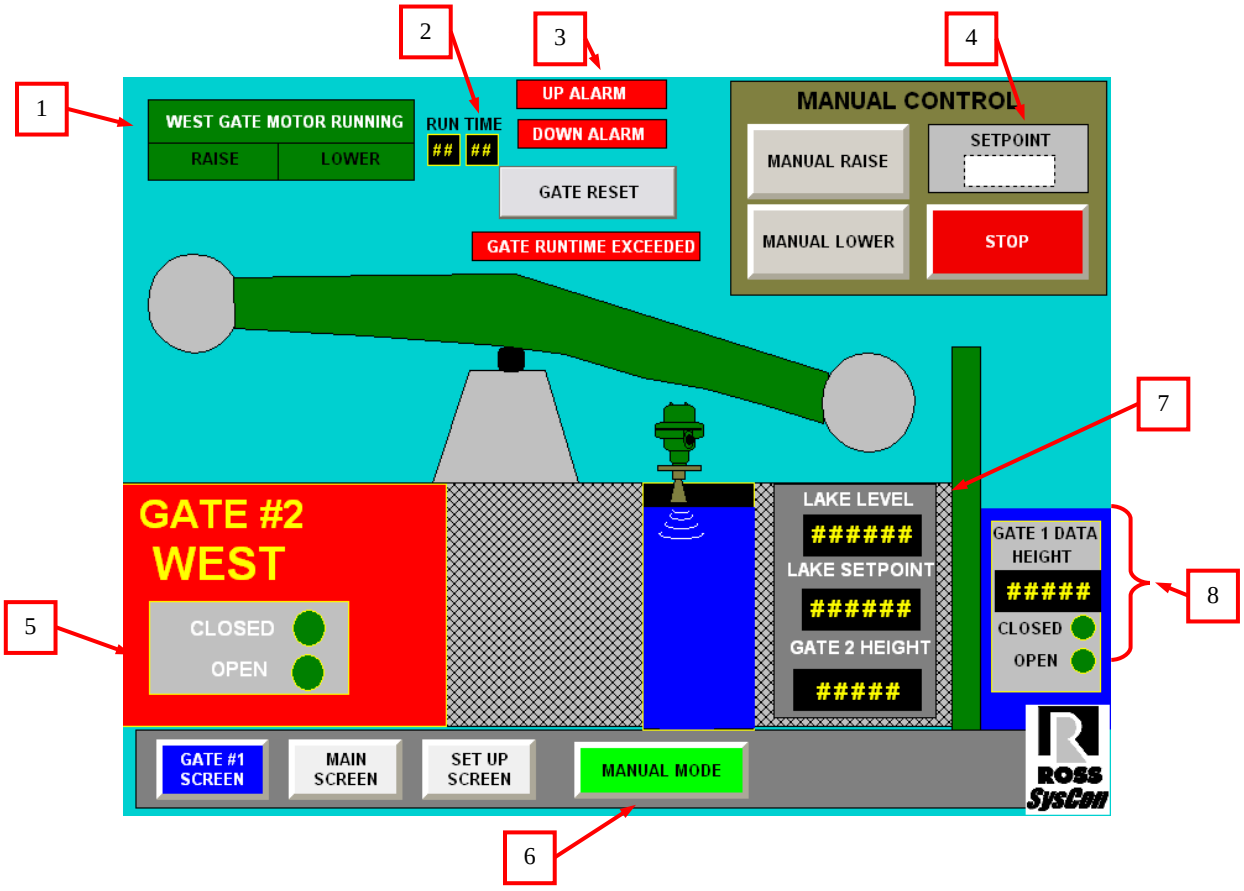
4). MANUAL CONTROL: ENTER SETPOINT HEIGHT AND PRESS ENTER. PRESS RAISE OR LOWER BUTTON TO OPERATE GATE. GATE WILL TRAVEL TO SETPOINT THEN STOP.

5). LIMIT SWITCH INDICATION: DISPLAYS STATUS OF GATE FULL DOWN LIMIT SWITCH.

6). MODE SWITCH: TOGGLE BUTTON TO CHANGE BETWEEN MANUAL AND AUTOMATIC MODES FOR GATE OPERATION.

7). LEVEL FIELDS: SHOWS LAKE LEVEL SETPOINT AND ACTUAL LEVEL (AS OF THE LAST CYCLE READING). DISPLAYS GATE HEIGHT FOR THE EAST GATE.

8). GATE 2 DATA: DISPLAYS GATE HEIGHT AND LIMIT SWITCH STATUS FOR WEST/GATE 2.



WEST/ GATE 2 SCREEN

1). **GATE DIRECTION INDICATION:** ILLUMINATES THE DIRECTION OF THE GATE WHEN IN RUN MODE.

2). **GATE RUN TIME:** DISPLAYS THE TOTAL AMOUNT OF RUN TIME FOR GATE 2. RUN TIME LIMIT IS SET ON CALIBRATION PAGE.

3). **GATE ALARMS:** INDICATORS APPEAR WHEN GATE IS IN ALARM STATE. PRESS RESET TO CLEAR ALARM.

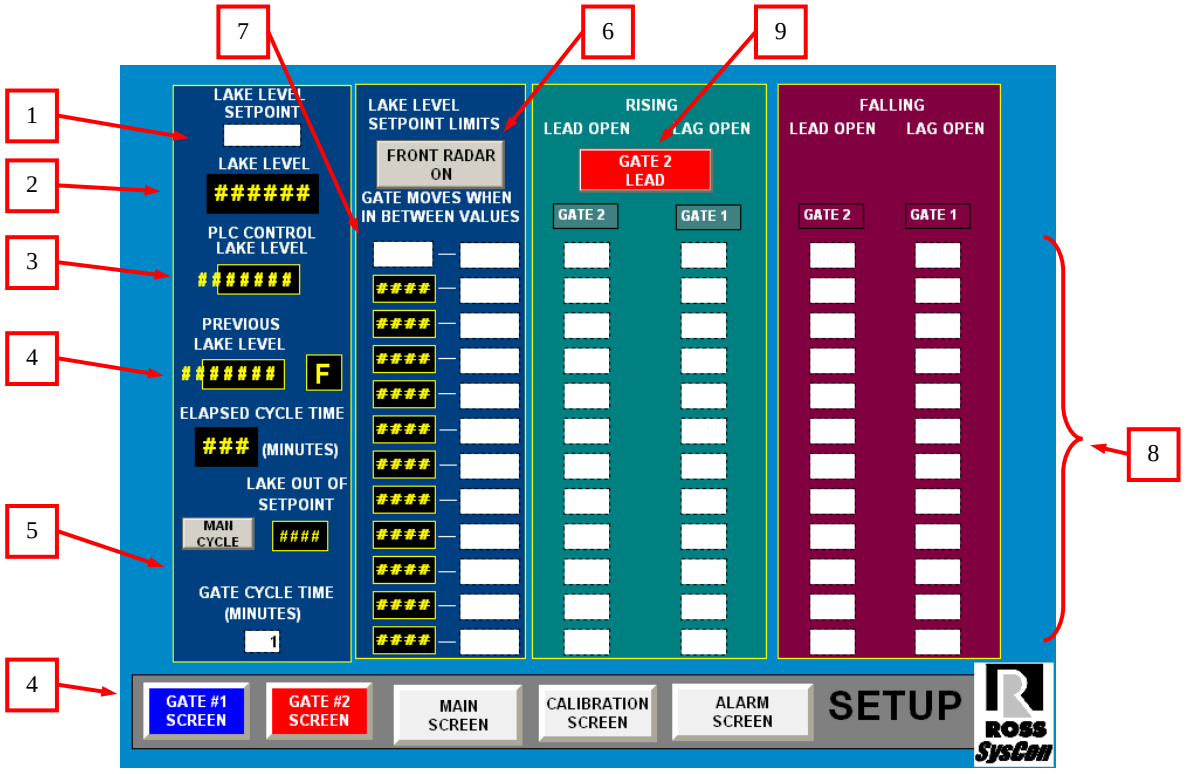
4). **MANUAL CONTROL:** ENTER SETPOINT HEIGHT AND PRESS ENTER. PRESS RAISE OR LOWER BUTTON TO OPERATE GATE. GATE WILL TRAVEL TO SETPOINT THEN STOP.

5). **LIMIT SWITCH INDICATION:** DISPLAYS STATUS OF GATE FULL DOWN LIMIT SWITCH.

6). **MODE SWITCH:** TOGGLE BUTTON TO CHANGE BETWEEN MANUAL AND AUTOMATIC MODES FOR GATE OPERATION.

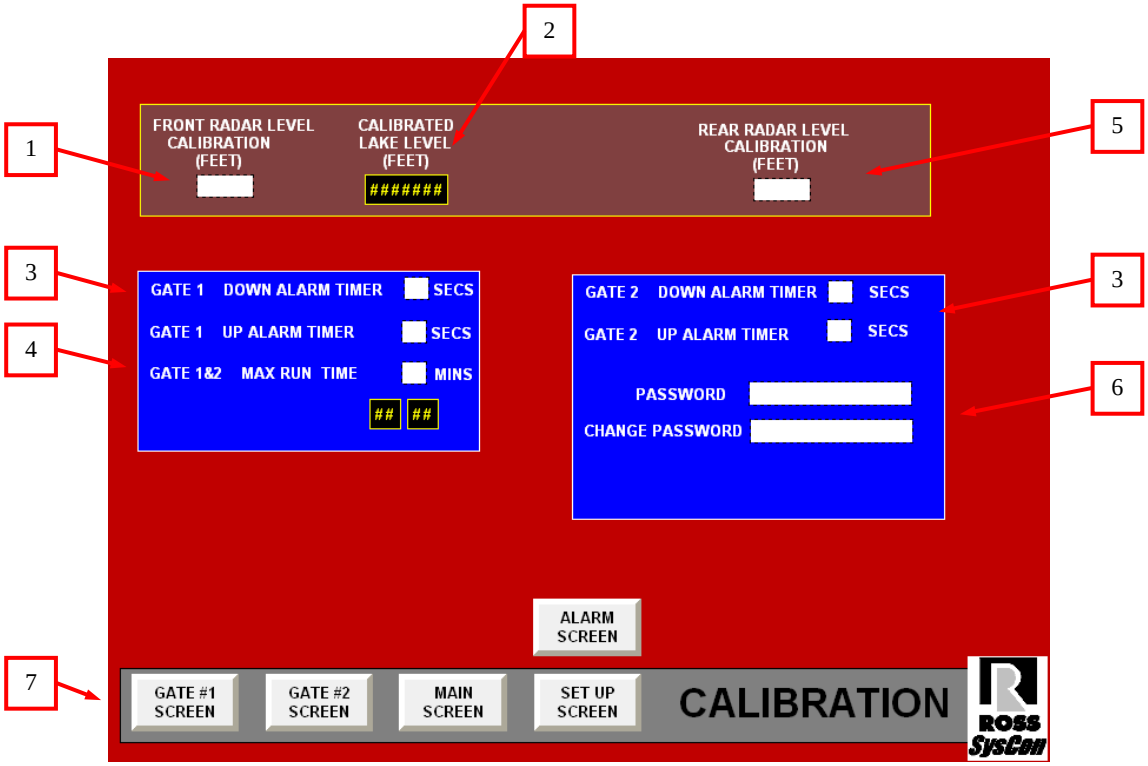
7). **LEVEL FIELDS:** SHOWS LAKE LEVEL SETPOINT AND ACTUAL LEVEL (AS OF THE LAST CYCLE READING). DISPLAYS GATE HEIGHT FOR THE EAST GATE.

8). **GATE 1 DATA:** DISPLAYS GATE HEIGHT AND LIMIT SWITCH STATUS FOR EAST/GATE 1.



SET UP SCREEN

1) **LAKE LEVEL SETPOINT:** ENTER LEVEL SETPOINT AND PRESS ENTER.
2) **ACTUAL LAKE LEVEL:** DISPLAYS ACTUAL LEVEL IN REAL TIME.
3) **PLC CONTROL LEVEL:** DISPLAYS LEVEL THAT PLC CONTROL IS CALCULATED FROM. CONTROL LEVEL IS UPDATED BASED ON THE GATE CYCLE TIME FIELD.
4) **PREVIOUS LAKE LEVEL:** DISPLAYS PREVIOUS LEVEL THAT WAS USED FOR CONTROL. THIS FIELD UPDATES WHEN PLC CONTROL LEVEL UPDATES. IT IS USED TO CALCULATE RISING OR FALLING LEVELS.
5) **GATE CYCLE CONTROL:** ENTER CYCLE TIME (MAX 200 MINUTES) AND PRESS ENTER. ELAPSED TIME DISPLAYS HOW LONG CYCLE HAS BEEN RUNNING. TO RESET CYCLE PRESS "**MAN CYCLE**" BUTTON. OUT OF SETPOINT FIELD SHOWS CURENT ERROR BETWEEN SETPOINT AND CONTROL LEVEL.
6) **RADAR SELECTION:** PRESS TO TOGGLE BETWEEN FRONT AND REAR RADAR LEVEL DETECTORS.
7) **OUT OF SETPOINT LIMIT FIELDS:** THESE FIELDS GIVE RANGES ABOVE LAKE LEVEL SET POINT AND CORRESPOND TO THE ADJACENT GATE HEIGHT SETPOINT FIELDS.
8) **GATE HEIGHT SETPOINT FIELDS:** THESE FIELDS SETUP THE AMOUNT OF MOVEMENT ON THE GATE PER OUT- OF SETPOINT RANGE. PRESS ENTER FOR COMPUTER TO ACCEPT DATA CHANGES.
9) **LEAD/LAG BUTTON:**THIS BUTTON SETS THE LEAD GATE FOR LEVEL CONTROL. PRESS TO TOGGLE BTWEEN EAST AND WEST GATE.



CALIBRATION SCREEN

1) **FRONT RADAR CAL NUMBER:** DISPLAYS CALIBRATION NUMBER USED FOR LAKE LEVEL CALCULATIONS FOR THE FRONT RADAR LEVEL DETECTOR. DETERMINE LAKE LEVEL BY SURVEY.
2) **LEVEL DISPLAY FIELD:** DISPLAYS CALIBRATED LAKE LEVEL.
3) **GATE ALARM TIMERS:** ENTER TIME SETPOINTS AND PRESS ENTER. WHEN GATE RUNS AND NO MOVEMENT IS MADE IN PRESET TIME, ALARM WILL STOP MOVEMENT.
4) **GATE RUNTIME SETPOINT:** ENTER AMOUNT OF TOTAL TIME GATE IS ALLOWED TO RUN BEFORE ALARMING. IF TIME EXPIRES, AND ALARM WILL APPEAR ON THE GATE SCREEN. PRESS THE RESET ON THE GATE SCREEN TO CLEAR ALARM. TIMER RESETS AT THE BEGINNING OF EVERY MOVEMENT CYCLE.
5) **REAR RADAR CAL NUMBER:** DISPLAYS CALIBRATION NUMBER USED FOR LAKE LEVEL CALCULATIONS FOR THE REAR RADAR LEVEL DETECTOR.
6) **PASSWORD FIELDS:** PASSWORD IS FOR SETTING MAX GATE RUNTIME. ENTER PASSWORD AND PRESS ENTER, GATE RUNTIME SETPOINT FIELD, AND CHANGE PASSWORD FIELD , WILL APPEAR. AFTER SETTING RUNTIME, CLEAR PASSWORD TO MAKE RUNTIME AND PASSWORD FIELDS INVISIBLE. TO CHANGE PASSWORD, TYPE IN NEW PASSWORD IN THE CHANGE PASSWORD FIELD.
7) **NAVIGATION BUTTONS:** ADVANCES OPERATOR TO THE SELECTED SCREEN.

SAFETY FEATURES

DOWN ALARMS: Set maximum run time on the down alarm so that the motor will not run longer than that once the gate height indicator stops indicating downward gate movement.

UP ALARMS: Set maximum run time on the up alarm so that the motor will not run longer than that once the gate height indicator stops indicating upward gate movement.

MAX RUN TIME: Set a maximum run time so that the gate will not run longer than a specified time. The motor will spool off all of the cable in approximately 25 minutes. Therefore, set a maximum run time to be much less than that, say 10 minutes. To start a new cycle if the motor reaches this limit

LIMIT SWITCH: Each gate has an upper and lower limit switch. The lower limit switch changes the gate height to 0.00 when it is engaged. The upper limit switch turns off the motor when the gate reaches the switch, at about 3 ft opening. When this limit switch is opened, the gate cannot be raised or lowered in auto mode, only in the "Hand" mode until the switch closes (after the gate goes below or above the switch.)

PLC: The PLC actually controls the dam, not the PC. The PC is the means to change the settings in the PLC, such as lake set point, lake level ranges, etc. If the computer fails, the PLC will continue to control the dam operation based on the last PC input.

CAUTIONS

COLD WEATHER: Cold weather causes the gear oil to become more viscous. Consider operating the gate manually on very cold mornings. Otherwise, an over current fault in the PLC may occur and the fault will have to be manually reset in the PLC.

